

7.1 Orthogonal Matrices

Chapter 7: Diagonalization and Quadratic Forms — Overview

Section 5.2 gave conditions for diagonalizability of $n \times n$ matrices without identifying which classes actually satisfy them. This chapter shows that *every symmetric matrix is diagonalizable* — an important result exploited across many applications.

In this section we discuss the class of matrices whose inverses can be obtained by transposition. Such matrices occur in a variety of applications and arise as transition matrices when one orthonormal basis is changed to another.

Orthogonal Matrices

Definition 1

A square matrix A is **orthogonal** if its transpose equals its inverse:

$$A^{-1} = A^T, \quad \text{or equivalently,} \quad AA^T = A^T A = I \quad (1)$$

A matrix transformation $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an **orthogonal transformation** (or **orthogonal operator**) if A is orthogonal.

EXAMPLE 1 | A 3×3 Orthogonal Matrix

The matrix

$$A = \begin{bmatrix} \frac{3}{7} & \frac{2}{7} & \frac{6}{7} \\ -\frac{6}{7} & \frac{3}{7} & \frac{2}{7} \\ \frac{2}{7} & \frac{6}{7} & -\frac{3}{7} \end{bmatrix}$$

is orthogonal since

$$A^T A = \begin{bmatrix} \frac{3}{7} & -\frac{6}{7} & \frac{2}{7} \\ \frac{2}{7} & \frac{3}{7} & \frac{6}{7} \\ \frac{6}{7} & \frac{2}{7} & -\frac{3}{7} \end{bmatrix} \begin{bmatrix} \frac{3}{7} & \frac{2}{7} & \frac{6}{7} \\ -\frac{6}{7} & \frac{3}{7} & \frac{2}{7} \\ \frac{2}{7} & \frac{6}{7} & -\frac{3}{7} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

EXAMPLE 2 | Rotation and Reflection Matrices Are Orthogonal

The standard matrix for the counterclockwise rotation of $\mathbb{R}^2$ through angle θ is

$$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

This matrix is orthogonal for all θ since

$$A^T A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The reflection matrices in Tables 1 and 2 of Section 1.8 are likewise all orthogonal (verify).

Both examples above share the property that their row and column vectors form orthonormal sets. This is a consequence of the following theorem.

Theorem 7.1.1

The following are equivalent for an $n \times n$ matrix A :

- (a) A is orthogonal.
- (b) The row vectors of A form an orthonormal set in $\mathbb{R}^n$ with the Euclidean inner product.
- (c) The column vectors of A form an orthonormal set in $\mathbb{R}^n$ with the Euclidean inner product.

Proof. We prove $(a) \Leftrightarrow (b)$; the proof of $(a) \Leftrightarrow (c)$ is analogous and left as an exercise.

Let $\mathbf{r}_i$ be the i th row vector and $\mathbf{c}_j$ the j th column vector of A . Since transposing swaps rows and columns, $\mathbf{c}_j^T = \mathbf{r}_j$. From the row-column rule and Table 1 of Section 3.2:

$$AA^T = \begin{bmatrix} \mathbf{r}_1 \cdot \mathbf{r}_1 & \mathbf{r}_1 \cdot \mathbf{r}_2 & \cdots & \mathbf{r}_1 \cdot \mathbf{r}_n \\ \mathbf{r}_2 \cdot \mathbf{r}_1 & \mathbf{r}_2 \cdot \mathbf{r}_2 & \cdots & \mathbf{r}_2 \cdot \mathbf{r}_n \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{r}_n \cdot \mathbf{r}_1 & \mathbf{r}_n \cdot \mathbf{r}_2 & \cdots & \mathbf{r}_n \cdot \mathbf{r}_n \end{bmatrix}$$

Thus $AA^T = I$ if and only if

$$\mathbf{r}_i \cdot \mathbf{r}_i = 1 \quad (i = 1, \dots, n) \quad \text{and} \quad \mathbf{r}_i \cdot \mathbf{r}_j = 0 \quad (i \neq j)$$

which holds if and only if $\{\mathbf{r}_1, \dots, \mathbf{r}_n\}$ is an orthonormal set in $\mathbb{R}^n$. ■

Warning. An orthogonal matrix has *orthonormal* rows and columns — not merely orthogonal.

Theorem 7.1.2

- (a) The transpose of an orthogonal matrix is orthogonal.
- (b) The inverse of an orthogonal matrix is orthogonal.
- (c) A product of orthogonal matrices is orthogonal.
- (d) If A is orthogonal, then $\det(A) = 1$ or $\det(A) = -1$.

The proofs are straightforward and are left as exercises.

EXAMPLE 3 | $\det(A) = \pm 1$ for an Orthogonal Matrix

The matrix

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

is orthogonal since its rows form an orthonormal set in $\mathbb{R}^2$. One can verify $\det(A) = 1$; interchanging the rows yields an orthogonal matrix with determinant -1 .

Properties of Orthogonal Transformations

Theorem 7.1.3

If A is an $n \times n$ matrix, the following are equivalent:

- (a) A is orthogonal.
- (b) $\|A\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x}$ in $\mathbb{R}^n$.
- (c) $A\mathbf{x} \cdot A\mathbf{y} = \mathbf{x} \cdot \mathbf{y}$ for all $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$.

Proof. We prove $(a) \Rightarrow (b) \Rightarrow (c) \Rightarrow (a)$.

$(a) \Rightarrow (b)$ Assume $A^T A = I$. By Formula (26) of Section 3.2:

$$\|A\mathbf{x}\| = (A\mathbf{x} \cdot A\mathbf{x})^{1/2} = (\mathbf{x} \cdot A^T A\mathbf{x})^{1/2} = (\mathbf{x} \cdot \mathbf{x})^{1/2} = \|\mathbf{x}\|$$

$(b) \Rightarrow (c)$ Assume $\|A\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x}$. By Theorem 3.2.7:

$$A\mathbf{x} \cdot A\mathbf{y} = \frac{1}{4}\|A\mathbf{x} + A\mathbf{y}\|^2 - \frac{1}{4}\|A\mathbf{x} - A\mathbf{y}\|^2 = \frac{1}{4}\|A(\mathbf{x} + \mathbf{y})\|^2 - \frac{1}{4}\|A(\mathbf{x} - \mathbf{y})\|^2 = \frac{1}{4}\|\mathbf{x} + \mathbf{y}\|^2 - \frac{1}{4}\|\mathbf{x} - \mathbf{y}\|^2 = \mathbf{x} \cdot \mathbf{y}$$

$(c) \Rightarrow (a)$ Assume $A\mathbf{x} \cdot A\mathbf{y} = \mathbf{x} \cdot \mathbf{y}$ for all $\mathbf{x}, \mathbf{y}$. By Formula (26) of Section 3.2,

$$\mathbf{x} \cdot \mathbf{y} = \mathbf{x} \cdot A^T A\mathbf{y} \implies \mathbf{x} \cdot (A^T A - I)\mathbf{y} = 0 \quad \text{for all } \mathbf{x}, \mathbf{y}.$$

Setting $\mathbf{x} = (A^T A - I)\mathbf{y}$:

$$(A^T A - I)\mathbf{y} \cdot (A^T A - I)\mathbf{y} = 0 \implies (A^T A - I)\mathbf{y} = \mathbf{0}$$

Since this holds for every $\mathbf{y}$, we have $A^T A - I = 0$, i.e., $A^T A = I$. ■

Parts (a) and (b) of Theorem 7.1.3 show that orthogonal operators are precisely those leaving norms (and hence distances and angles) unchanged — as illustrated in Figure 7.1.1.

Change of Orthonormal Basis

Orthonormal bases are convenient because the following theorem shows that familiar formulas continue to hold for norms, distances, and inner products when coordinates are computed relative to such a basis.

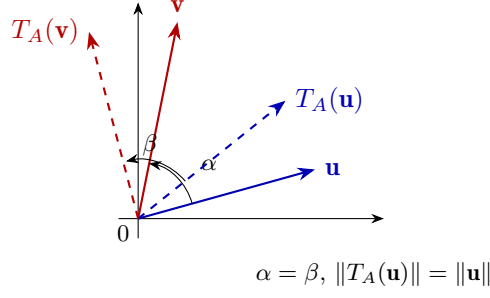


Figure 1: Orthogonal operator T_A preserves norms and angles: $\beta = \alpha$.

Theorem 7.1.4

If S is an orthonormal basis for an n -dimensional inner product space V , and

$$(\mathbf{u})_S = (u_1, u_2, \dots, u_n), \quad (\mathbf{v})_S = (v_1, v_2, \dots, v_n)$$

then:

- (a) $\|\mathbf{u}\| = \sqrt{u_1^2 + u_2^2 + \dots + u_n^2}$
- (b) $d(\mathbf{u}, \mathbf{v}) = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \dots + (u_n - v_n)^2}$
- (c) $\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$

Remark. The three parts can be restated as

$$\|\mathbf{u}\| = \|(\mathbf{u})_S\|, \quad d(\mathbf{u}, \mathbf{v}) = d((\mathbf{u})_S, (\mathbf{v})_S), \quad \langle \mathbf{u}, \mathbf{v} \rangle = \langle (\mathbf{u})_S, (\mathbf{v})_S \rangle$$

so norms, distances, and inner products in V equal those of the coordinate vectors in $\mathbb{R}^n$ relative to any orthonormal basis.

Transitions between orthonormal bases for inner product spaces are of special importance in geometry and applications. The following theorem characterizes the transition matrix.

Theorem 7.1.5

Let V be a finite-dimensional inner product space. If P is the transition matrix from one orthonormal basis B' to another orthonormal basis B for V , then P is an orthogonal matrix.

Proof. Let $\|\cdot\|_V$ denote the norm on V and $\|\cdot\|$ the Euclidean norm on $\mathbb{R}^n$. Since B' is an orthonormal basis, Theorem 7.1.4(a) gives for any $\mathbf{u} \in V$:

$$\|\mathbf{u}\|_V = \|[\mathbf{u}]_{B'}\| \tag{6}$$

Let $\mathbf{x}$ be any vector in $\mathbb{R}^n$ and $\mathbf{u}$ the vector in V with $[\mathbf{u}]_{B'} = \mathbf{x}$, so $P\mathbf{x} = [\mathbf{u}]_B$. Applying (6) to both bases:

$$\|\mathbf{u}\|_V = \|[\mathbf{u}]_{B'}\| = \|P\mathbf{x}\| \quad \text{and} \quad \|\mathbf{u}\|_V = \|[\mathbf{u}]_B\| = \|\mathbf{x}\|$$

Therefore $\|P\mathbf{x}\| = \|\mathbf{x}\|$ for every $\mathbf{x} \in \mathbb{R}^n$, so P is orthogonal by Theorem 7.1.3. ■

Rotation Equations

EXAMPLE 4 | Rotation of Axes in 2-Space

When a rectangular xy -coordinate system is rotated counterclockwise through an angle θ about the origin to produce an $x'y'$ -system, each point Q acquires two sets of coordinates. Introducing unit vectors $\mathbf{u}_1, \mathbf{u}_2$ along the x -, y -axes and $\mathbf{u}'_1, \mathbf{u}'_2$ along the x' -, y' -axes, this is a basis change from $B = \{\mathbf{u}_1, \mathbf{u}_2\}$ to $B' = \{\mathbf{u}'_1, \mathbf{u}'_2\}$. Using Formulas (7)–(8) of Section 4.7 one deduces

$$[\mathbf{u}_1]_{B'} = \begin{bmatrix} \cos \theta \\ -\sin \theta \end{bmatrix}, \quad [\mathbf{u}_2]_{B'} = \begin{bmatrix} \sin \theta \\ \cos \theta \end{bmatrix}$$

so the transition matrix and the **rotation equations** are

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = P \begin{bmatrix} x \\ y \end{bmatrix}, \quad P = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \quad (4)$$

or equivalently,

$$\boxed{x' = x \cos \theta + y \sin \theta, \quad y' = -x \sin \theta + y \cos \theta} \quad (5)$$

By Theorem 7.1.5, P is orthogonal, as can be verified directly.

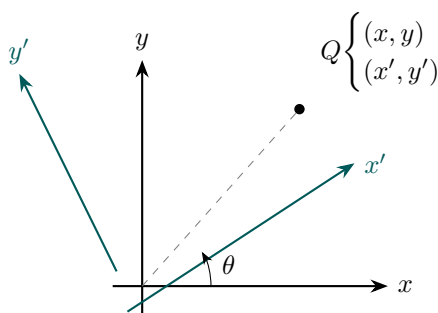


Figure 2: Rotation of the xy -axes through angle θ giving $x'y'$ -coordinates.

EXAMPLE 5 | Rotation of Axes in 2-Space

SOLUTION: Find the new coordinates of $Q(2, -1)$ after rotating the axes through $\theta = \pi/4$. Since $\sin(\pi/4) = \cos(\pi/4) = 1/\sqrt{2}$, equation (4) gives:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{3}{\sqrt{2}} \end{bmatrix}$$

So the new coordinates of Q are $(\frac{1}{\sqrt{2}}, -\frac{3}{\sqrt{2}})$.

Remark. The coefficient matrix in (4) is the standard matrix for the linear operator that rotates vectors through $-\theta$. Rotating the coordinate axes through θ with vectors fixed has the same effect as rotating the vectors through $-\theta$ with the axes fixed.

EXAMPLE 6 | Rotation of Axes in 3-Space

Suppose a rectangular xyz -system is rotated counterclockwise (looking down the positive z -axis) through angle θ about the z -axis to produce an $x'y'z'$ -system. With unit vectors $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ along the xyz -axes

and $\mathbf{u}'_1, \mathbf{u}'_2, \mathbf{u}'_3$ along $x'y'z'$ -axes:

$$[\mathbf{u}'_1]_B = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 0 \end{bmatrix}, \quad [\mathbf{u}'_2]_B = \begin{bmatrix} -\sin \theta \\ \cos \theta \\ 0 \end{bmatrix}, \quad [\mathbf{u}'_3]_B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

The transition matrix from B' to B and its inverse are

$$P = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The new coordinates of a point Q with old coordinates (x, y, z) are:

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

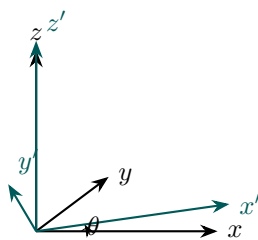


Figure 3: Rotation of the xyz -axes through angle θ about the z -axis.

Exercise Set 7.1

In Exercises 1–4, determine whether the matrix is orthogonal, and if so find its inverse.

1. **a.** $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ **b.** $\begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$

3. **a.** $\begin{bmatrix} 0 & 1 & \frac{1}{\sqrt{2}} \\ 1 & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$ **b.** $\begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$

In Exercises 5–6, show that the matrix is orthogonal three ways: by computing $A^T A$, by using Theorem 7.1.1(b), and by using Theorem 7.1.1(c).

5. $A = \begin{bmatrix} \frac{4}{5} & 0 & -\frac{3}{5} \\ -\frac{9}{25} & \frac{4}{5} & -\frac{12}{25} \\ \frac{12}{25} & \frac{3}{5} & \frac{16}{25} \end{bmatrix}$

7. Let $T_A: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be multiplication by the orthogonal matrix in Exercise 5. Find $T_A(\mathbf{x})$ for $\mathbf{x} = (-2, 3, 5)$, and confirm that $\|T_A(\mathbf{x})\| = \|\mathbf{x}\|$ with the Euclidean inner product on $\mathbb{R}^3$.

9. Are the standard matrices for the reflections in Tables 1 and 2 of Section 1.8 orthogonal?
11. What conditions must a and b satisfy for the matrix $\begin{bmatrix} a+b & b-a \\ a-b & b+a \end{bmatrix}$ to be orthogonal?
12. Under what conditions will a diagonal matrix be orthogonal?
- Exercises 13–14: find coordinates after rotation of axes.*
13. A rectangular $x'y'$ -system is obtained by rotating the xy -system counterclockwise through $\theta = \pi/3$.
- Find the $x'y'$ -coordinates of the point with xy -coordinates $(-2, 6)$.
 - Find the xy -coordinates of the point with $x'y'$ -coordinates $(5, 2)$.
15. A rectangular $x'y'z'$ -system is obtained by rotating the xyz -system counterclockwise about the z -axis through $\theta = \pi/4$.
- Find the $x'y'z'$ -coordinates of the point with xyz -coordinates $(-1, 2, 5)$.
 - Find the xyz -coordinates of the point with $x'y'z'$ -coordinates $(1, 6, -3)$.
21. A linear operator on $\mathbb{R}^2$ is called *rigid* if it does not change lengths of vectors, and *angle preserving* if it does not change the angle between nonzero vectors.
- Identify two different types of linear operators on $\mathbb{R}^2$ that are rigid.
 - Identify two different types that are angle preserving.
 - Are there any linear operators on $\mathbb{R}^2$ that are rigid but not angle preserving? Justify your answer.
24. The sets $S = \{1, x\}$ and $S' = \{\frac{1}{\sqrt{2}}(1+x), \frac{1}{\sqrt{2}}(1-x)\}$ are orthonormal bases for P_1 with the standard inner product. Find the transition matrix P from S to S' , and verify that the conclusion of Theorem 7.1.5 holds for P .
25. Prove that if $\mathbf{x}$ is an $n \times 1$ matrix, then

$$A = I_n - \frac{2}{\mathbf{x}^T \mathbf{x}} \mathbf{x} \mathbf{x}^T$$

is both orthogonal and symmetric.

26. Prove that a 2×2 orthogonal matrix A has exactly two possible forms:

$$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad \text{or} \quad A = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}$$

for some $0 \leq \theta < 2\pi$. [*Hint:* Start with a general 2×2 matrix A and use the fact that the column vectors form an orthonormal set in $\mathbb{R}^2$.]

30. Prove the equivalence of statements (a) and (c) in Theorem 7.1.1.